COMPUTER ENGINEERING

QUEZON CITY

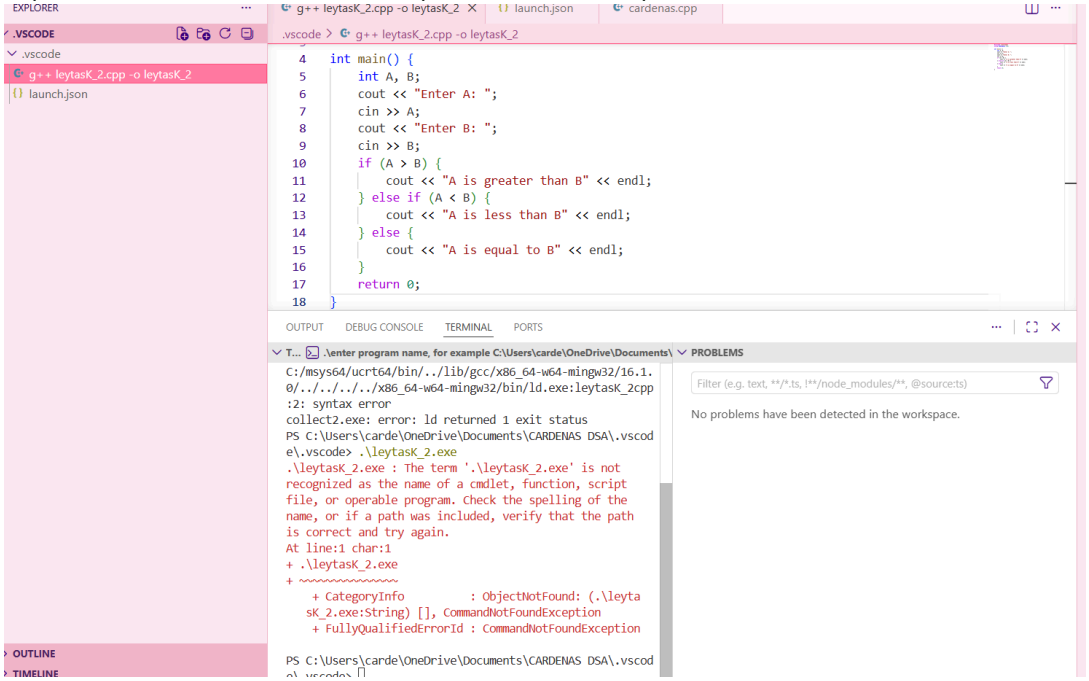
COURSE CODE/TITLE:	CPE 0120	**SECTION:**	CPE21S4
DATE PERFORMED:	JULY 14 2026	**DATE SUBMITTED:**	JULY 14 2026
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ACTIVITY NO. 1.1 Basic C++ Programming

1. PROCEDURE OUTPUT

What procedures did you perform?

The program declares two integer variables, A and B, to store user inputs. It prompts the user to enter the first number into A and the second number into B. Then, it compares these values using an if-else if-else statement. If A is greater than B, it displays "A is greater than B." If A is less than B, it shows "A is less than B." If both are equal, it outputs "A is equal to B." After displaying the result, the program terminates successfully. This simple process helps the user determine the relationship between the two input numbers.



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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

1. Create a C++ to swap the two numbers in different variables.
    - This C++ program exchanges the values of two variables using a temporary variable. After the swap, each variable contains the other's original value.

```
10    int main() {
18
19        swapNumbers(x, y);
20
21        cout << "After swapping:" << endl;
22        cout << "First number: " << x << endl;
23        cout << "Second number: " << y << endl;
```

```
OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
TERMI... .\enter program name, for example C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\a.exe
-a---l        15/07/2026  12:26 am            499 rietasK_.cpp


PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> pwd

Path
----
C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode


PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> g++ rietasK_.cpp
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> g++ rietask_.cpp -o rieta
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> .\rietask.exe
Enter first number: 29
Enter second number: 31
After swapping:
First number: 31
Second number: 29
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode>
```

2. Create a C++ program that has a function to convert temperature in kelvin to fahrenheit.

   - Using a function, the program successfully changed the temperature from Kelvin to Fahrenheit. It accurately displayed the equivalent temperature in Fahrenheit after accepting the user-entered Kelvin value and applying the proper conversion formula. The program became more structured, reusable, and comprehensible by utilizing a function.

```
5   double kelvinToFahrenheit(double kelvin)
7       return (kelvin - 273.15) * 9.0 / 5.0 + 32.0;
8   }
9   int main()
10  {
11      double kelvin;
12      cout << "Enter temperature in Kelvin: ";
13      cin >> kelvin;
14
15      double fahrenheit = kelvinToFahrenheit(kelvin);
16      cout << kelvin << " K is equal to " << fahrenheit << " °F" << endl;
17
18      return 0;
```

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
TERMI... .\enter program name, for example C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\a.exe
    + FullyQualifiedErrorId : CommandNotFoundException

Suggestion [3,General]: The command \tasK_2.exe was not found, but does exist in the current
n. Windows PowerShell does not load commands from the current location by default. If you trus
 command, instead type: ".\\tasK_2.exe". See "get-help about_Command_Precedence" for more deta
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> .\task_2.exe
Enter temperature in Kelvin: 500
500 K is equal to 440.33 °F
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> .\task_2.exe
Enter temperature in Kelvin: 300
300 K is equal to 80.33 °F
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode>
```

3. Create a C++ program that has a function that will calculate the distance between two points.

   - This program calculates the distance between two points on a coordinate plane. It uses a function that accepts the coordinates of the two points, applies the distance formula, and returns the calculated distance. This demonstrates the use of functions to perform mathematical calculations in C++.

```
4
5   // Function to calculate the distance between two points
6   double calculateDistance(double x1, double y1, double x2, double y2)
7   {
8       return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
9   }
10
11  int main()
12  {
13      double x1, y1, x2, y2, distance;
14
15      cout << "Enter the coordinates of the first point (x1 y1): ";
16      cin >> x1 >> y1;
17
18      cout << "Enter the coordinates of the second point (x2 y2): ";
19      cin >> x2 >> y2;
20
21      distance = calculateDistance(x1, y1, x2, y2);
22
23      cout << "The distance between the two points is: " << distance << endl;
24
25      return 0;
```

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
TERMI... .\enter program name, for example C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\a.exe

PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> .\task_2.exe
Enter temperature in Kelvin: 300
300 K is equal to 80.33 °F
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> g++ tasK_3.cpp
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> g++ task_3.cpp -o tasK_3
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> .\tasK_3.exe
Enter the coordinates of the first point (x1 y1): 7
8
Enter the coordinates of the second point (x2 y2): 10
11
The distance between the two points is: 4.24264
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode>
```

4. Modify the code given to compute given in ILO B and add the following functions:

    a . function to compute for the area of a triangle.
    B. function to compute for the perimeter of triangle
    C. function that determines whether the triangle is acute, obtuse or others angles

- This modified program adds functions to compute the area and perimeter of a triangle and determine whether the triangle is acute-angled, obtuse-angled, or another type based on its angles. Each function performs a specific task, making the program more organized and easier to understand. This activity demonstrates the use of object-oriented programming and functions to solve multiple problems within a single C++ program.

```
22  void Triangle::setAngles(double A, double B, double C) {
26      totalAngle = A + B + C;
27  }
28
29  bool Triangle::validateTriangle() const {
30      return (totalAngle == 180);
31  }
32
33  int main() {
34      Triangle set1(40, 30, 110);
35
36      if (set1.validateTriangle()) {
37          cout << "The shape is a valid triangle.\n";
38      } else {
39          cout << "The shape is NOT a valid triangle.\n";
40      }
```

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
TERMI... .\enter program name, for example C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\a.exe
The distance between the two points is: 4.24264
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> "enter program name, for exam
C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode/a.exe"
enter program name, for example C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode/a.exe
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> g++ tasK_4.cpp
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> g++ tasK_4.cpp -o tasK_4
PS C:\Users\carde\OneDrive\Documents\CARDENAS DSA\.vscode\.vscode> .\tasK_4.exe
Enter the three angles of the triangle: 40
30
110
The angles form a valid triangle.
```

Conclusion:
This exercise helped me learn how to write programs using functions, classes, and objects as well as the fundamental structure of C++ programming. Function prototypes, constructors, member functions, and object-oriented programming ideas like encapsulation became clearer to me. Additionally, I learned how to create objects, define classes, and use C++ to solve programming problems.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Grammarly and Quillbot**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Correction of grammar and more sentence to my activity by Quillbot extensions**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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